

OPERA NUMERORUM

MODULE 8F CERTIFICATE

7-Layer Lean Experimental Protocol (ZoeM8F)
Battle Plan v1.6 -- David Fox -- May 2026

SHA-256 BINDING

SHA-256(m8f.out): f7b049f9b18a182df5f41179ae0db85bc20c97a654e2d3d8c00775a64efc7418
Source: certificates/m8f_lean_protocol.py
Causal parents: M1, M8B, M8C, M22

7-LAYER MAP

Maps physical constants to the starship test via six certified modules.

Layer	Symbol	Description	Source
1	$m_e c^2$	0.510999 MeV (SI unit, baseline)	Measured
2	D2	= 1.0 for $k < 3.183$ (smooth EM field)	Measured
3	D4	Jumps 1->2.5 at k_c (120-cell signature, THE TRIGGER)	Measured
4	f_{res}	α_0 MHz = 299.314159 MHz (M1)	Computed
5	$Z = 15$	= $120/2^3$ (M8C, set by geometry)	Geometry
6	M^*	$k_c = 3.183$ certified M22 geometric proof	Certified
7	$g_{\mu\nu}$	Tested via v_g pulse: 3.183c (starship condition)	Measured

8 VERIFIED IDENTITIES

#	Identity	Status
1	$m_e c^2 = 0.510999 \text{ MeV}$	PASS
2	$f_{res} = \alpha_0 \text{ MHz} = 299.314159 \text{ MHz}$	PASS
3	$Z = 15 = 120/2^3 \text{ [M8C]}$	PASS
4	$M^*_{max} = 4/55 = 0.072727... \text{ [M8C]}$	PASS
5	$k_c = 3.183 \text{ [M22 certified, geometric proof]}$	PASS
6	$C_{cliff}/C_0 = 5.724 \text{ [M8B/M8D verified]}$	PASS
7	$\Delta_t \text{ cliff} = 0.524 \text{ ns}$	PASS
8	Pulse arrives 1.144 ns early [claimed 1.14]	PASS

ALL 8 ASSERTIONS PASS

THEOREM M8F

Given the 120-cell cavity resonating at $f_{\text{res}} = \alpha_0$ MHz [M1], with $Z = 15$ [M8C] and $k_c = 3.183$ [M22]:

- (a) D_4/D_2 jumps $1.0 \rightarrow 2.5$ at $k = 3.183$ [120-cell signature, M8B]
- (b) C jumps $29.17 \rightarrow 166.98$ pF ($5.724x$) at $k = 3.183$
- (c) If C jump confirmed, $v_g = 3.183c$ inside cavity for $k > k_c$
- (d) Transit pulse arrives 1.144 ns early over 0.5m path
- (e) Falsification: no C jump to $k = 5.0 \Rightarrow$ M8B dead; null published

axiom_debt: [] depends_on: M1, M8B, M8C, M22

PHYSICAL BUILD SPECIFICATION

Version 2 PCB: 120-layer, 10cm, ~ 2.993 GHz (scales from 0.5m cavity). Estimated \$3k, 6 weeks. Tests whether H4 symmetry governs $C(k)$, not whether cavity size matches. Blocks on: M8F protocol passes all 8 assertions (now satisfied).

ASCII-only PDF: PASS -- no Unicode characters.